

# Parallel Connection of GaN FETs: an Experimental Investigation Approach

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## Abstract

The paper investigates the parallel connection of the GaN FETs in inverters for low-voltage motor control applications. Several simulation results and experimental tests show the main device parameters and layout constraints involved in the effectiveness of parallel connection arrangement. A dedicated measurement board setup has been developed for the experimental evaluation. The experimental tests enable the empirical prediction of the application's maximum phase load current amplitude, considering the effects of the spread of the parameters among the parallel-connected GaN FETs. Furthermore, an inverter board with four devices in parallel is designed and assembled to validate the advanced GaN FETs parallel operation in an actual application, increasing the output phase current value for high-current motor drive applications.

## 1 Introduction

Nowadays, power converter applications demand increasing amounts of power in small volumes. In high-current, low-voltage applications such as motor drives for mild hybrid automotive applications, high-performance robotic systems, or Unmanned Aerial Vehicle (UAV) [1-3], the Gallium Nitride devices (GaN FET) are more attractive compared to Si MOSFET devices. The GaN FETs are chosen because of their high-performance characteristics, such as high-speed switching and low conduction losses. This makes them suitable for motor control applications [4], [5]. For thermal management purposes, using power devices in parallel is worthwhile, especially with insulated gate devices with a positive temperature coefficient that allows a current balance in steady state [6].

In low-voltage applications (less than 200V), connecting multiple devices in parallel configurations becomes essential to expand the utilization of GaN FETs in high-power applications. However, the parallel connection of devices is challenging for the PCB design phase and the devices' parametric spread management [7]. Parameter mismatch among the parallel circuit legs significantly affects the dynamic characteristics and thermal management of GaN FETs. To mitigate the influence of these parasitic components, alternative packaging techniques for power devices have been proposed [8].

The main objective of this paper is to assess the feasibility and effectiveness of using GaN FETs in parallel within an inverter setup to control low-voltage AC motors. The paper examines device parameters and layout constraints that impact the effectiveness of parallel connections. This could include the number of devices to connect in parallel, gate driving control, thermal management, and circuit layout considerations.

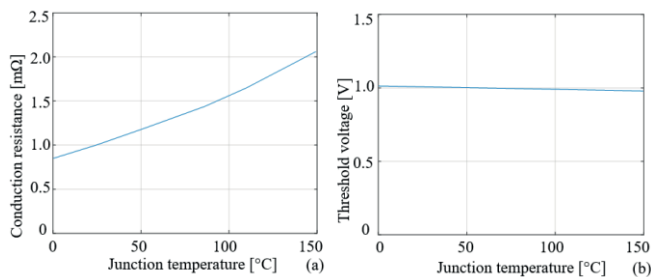
The paper involves both simulation and experimental testing. Simulation results are used to understand how GaN FETs behave in parallel configurations, predicting and making them better understand the outcomes before experimental tests in actual operative conditions. The experimental evaluation setup is designed to provide accurate and controlled conditions for testing the GaN FETs in parallel configuration and to validate the simulation results. Furthermore, the paper discusses an inverter circuit with four paralleled GaN FETs to demonstrate how GaN FETs can be used in parallel to increase the output phase current, which is crucial for high-current motor drive applications.

## 2 Design considerations of parallel GaN FETs and simulation results

Parallel connections of GaN FET devices demand careful consideration of various static and dynamic factors to achieve uniform current sharing and reliable operation. Several factors impact device temperature differences. The junction temperature variation may result in dynamic or static current-sharing unbalances depending on device characteristics. The behavior of the conduction resistance  $R_{DS(on)}$  and the threshold voltage  $V_{GSth}$  of the GaN FET as a function of the junction temperature is shown in Fig.1a and Fig.1b, respectively.

The parameters' spread plays a crucial role in the parallel connection of devices, both during conduction and switching transients [9].

In GaN FETs' parallel connections,  $R_{DS(on)}$  significantly affects static current sharing. However, the positive temperature coefficient aids current self-balancing. On the other hand, dynamic current sharing during turn-on and turn-off is mainly influenced by the spread of  $V_{GSth}$ . The lower threshold voltage  $V_{GSth}$  leads to an earlier turn-on, higher switching current/loss, and positive feedback [10]. Also, the transconductance ( $g_m$ ) plays a role in dynamic current sharing.



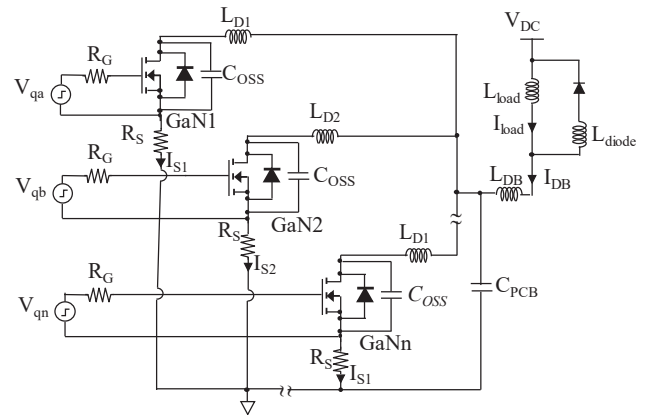
**Fig. 1** Dependency of GaN FET parameters on junction temperature: a) conduction resistance, b) threshold voltage.

A balanced circuit layout is crucial for dynamic current sharing and parallel operation stability. This is especially critical for high-speed power switches like GaN/SiC. In addition to synchronized device control, layout symmetry is pivotal.

Simulations are carried out to understand the impact of the  $V_{GSth}$  spread and the parasitic components of the circuit layout when several GaN FETs are connected in parallel. Fig.2 shows the electrical schematic of the system composed of  $N$  GaN FETs in parallel connection and connected to an inductive load with a freewheeling diode in parallel.

In Fig. 2, the GaN FETs gates are driven by a pulse voltage source  $V_q=5$  V, with a series gate resistance of  $1\ \Omega$ . Eight ( $N=8$ ) GaN FETs (EPC2302) are connected in parallel. In the simulation test circuit shown in Fig.2, the devices are arranged in rows considering the equivalent parasitic inductances for each GaN FET. The output capacitance  $C_{OSS} = C_{DS} + C_{GD}$  is represented in parallel to each GaN FET. The GaN FET model is described in [11].  $L_D=0.3$  nH represents the drain inductance due to the parallel connection. Shunt resistors  $R_S=100$  mΩ are used for sensing the GaN FET currents. The equivalent capacitance of the PCB is  $C_{PCB}=11$  nF. The bus voltage is  $V_{DC}=24$  V, and the load is  $L_{Load}=90\ \mu$ H. The freewheeling diode has a parasitic inductance of  $L_{diode}=1$  nH. Moreover, the load is connected to the GaN FETs through the drain-board inductance  $L_{DB}=1$  nH.

These stray inductances and  $C_{PCB}$  values come from actual measurements of a PCB mounting twelve GaN FETs in parallel.

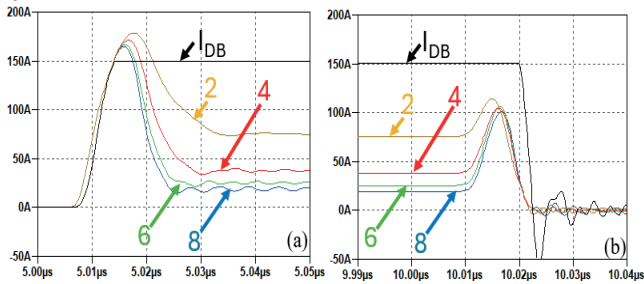


**Fig. 2**  $N$  GaN FETs in parallel connected to an inductive load with a freewheeling diode in parallel.

Different simulations are carried out by turning on only 2, 4, 6, or 8 GaN FETs at a time while the others remain in the off-state. The load current is  $I_{Load}=150$  A. Only the single GaN FET that turns on has a low-value threshold voltage,  $V_{th\ low}=0.95$  V. The other GaN FETs have a higher threshold voltage of  $V_{th\ high}=1.65$  V. The device's current with  $V_{th\ low}$  is shown in Fig. 3a for the turn-on and in Fig. 3b for the turn-off. The current waveforms in the cases of turning on 2, 4, 6, or 8 GaN FETs are indicated with the respective number. The current  $I_{DB}$  entering the common drain of the GaN FETs in parallel is depicted in black.

After the transient, during the conduction time, the  $I_{Load}$  is equally shared between the GaN FETs in the on-state. A higher number of devices in the on-

state results in a lower current conducted by each GaN FET because the simulation always used the same load current. The different threshold voltage causes the device with  $V_{th\ low}$  to switch on earlier and switch off later than the other GaN FETs with typical  $V_{th}$ .



**Fig. 3** Current of the GaN FET with  $V_{th\ low}$ . Number of GaN FETs turning on: 2 (yellow), 4 (red), 6 (green), 8 (blue). The current  $I_{DB}$  entering the common drain of the GaN FETs in parallel is shown in black. a) turn-on. b) turn-off.  $I=50\text{ A/div}$ . Time scale =  $10\text{ ns/div}$ .

At turn-on and turn-off, the current peak is higher if the GaN FET has  $V_{th\ low}$ . In this simulation,  $I_{DB}$  rises to the  $I_{load}$  value at turn-on and falls to 0 A quickly because of the low  $L_{diode}$  and  $L_{DB}$  values.

At turn-on, the GaN FET with  $V_{th\ low}$  conducts  $I_{load}$ . Moreover, the current shows a peak higher than  $I_{load}$  due to the discharging parasitic capacitances in the circuit,  $C_{OSS}$ , and  $C_{PCB}$ .

Similarly, during the turn-off, the GaN FET with  $V_{th\ low}$  has a peak lower than that at turn-on. For both turn-on and turn-off, the peak is higher in the case of only two GaN FETs turning on because of the higher current conducted at steady state.

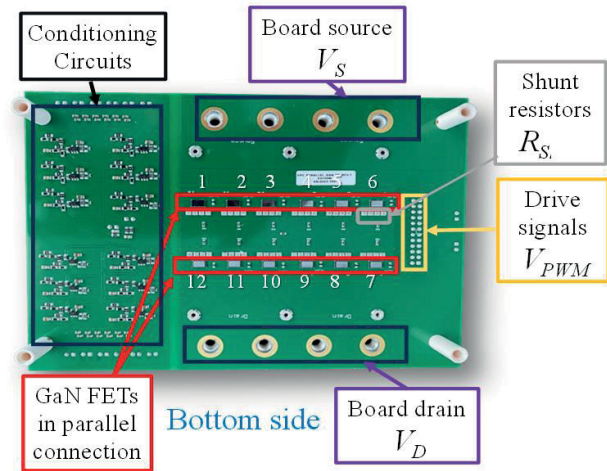
To separate the current peak due to the parasitic capacitance from the load current ( $I_{DB}$ ) rise, it is necessary to slow down the  $I_{DB}$  dynamic by  $L_{diode}$  and  $L_{DB}$  with higher values than the interconnecting inductances.

### 3 PCB with eight GaN FETs in parallel

The experimental evaluation is carried out with the board prototype equipped with up to twelve GaN FETs in parallel. The board is shown in Fig.4.

Each GaN FET has an independent gate driver in the prototype to decide how many devices supply the load. Moreover, a turn-on and turn-off mismatch between the GaN FETs can be set by adding a proper delay on the different control signals  $V_{PWM}$  applied to the GaN FETs. The control signal

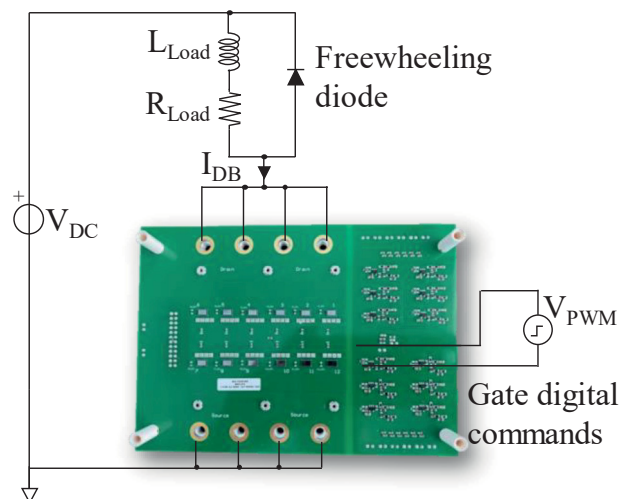
delay allows us to test and emulate the different operative conditions and  $V_{GStH}$  spread. Shunt resistors  $R_S=0.1\ \Omega$  are used to sense each GaN FET current.



**Fig. 4** Picture of the experimental parallel-GaN FETs test board.

The experimental test is carried out by connecting the board as depicted in Fig.5. The load is composed by  $L_{Load}=90\ \mu\text{H}$  and  $R_{Load}=0.1\ \Omega$  connected in parallel with a freewheeling diode. The DC voltage is  $V_{DC}=24\text{ V}$ .

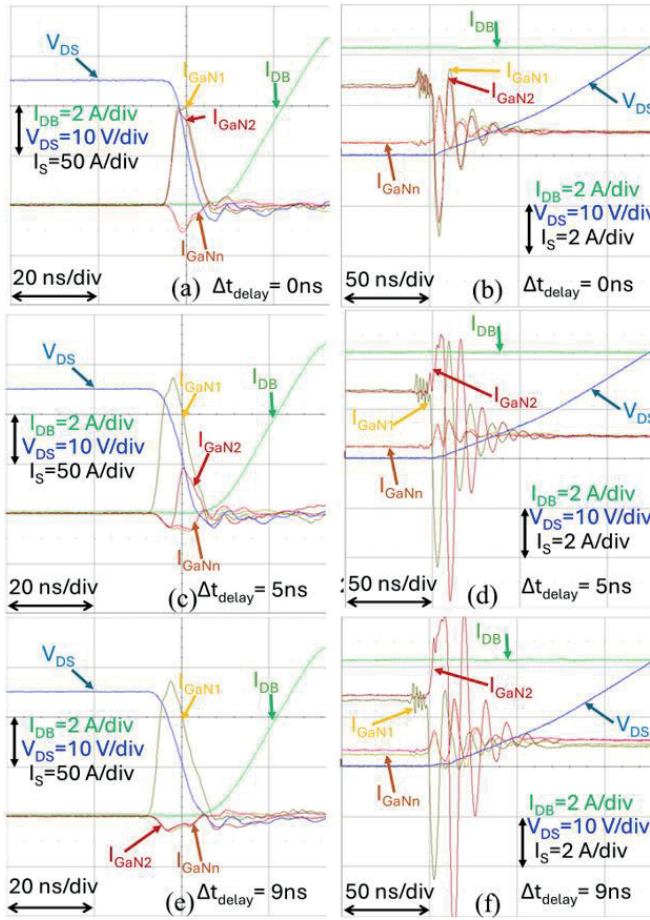
**Fig. 5** Electric schematic of the connection of the



parallel-GaN FETs board for the experimental tests.

In the experimental tests, two GaN FETs are driven by two  $V_{PWM}$  with an established delay between them. The gate driving of  $GaN_2$  is delayed with respect to  $GaN_1$ . All the other GaN FETs ( $GaN_n$ ) remain in off-state.

Fig. 6 depicts the waveforms of the GaN FETs' current  $I_S$ , the current  $I_{DB}$  entering the board, and the drain-source voltage  $V_{DS}$  of the GaN FETs in parallel, which are measured during the turn-on and turn-off transients.



**Fig. 6** Switching waveforms with GaN<sub>1</sub> driven earlier than GaN<sub>2</sub> that is delayed. a) turn-on delay=0 ns. b) turn-off delay=0 ns. c) turn-on delay=5 ns. d) turn-off delay=5 ns. e) turn-on delay=9 ns. f) turn-off delay=9 ns.

The switching waveforms obtained with delays of 0 ns, 5 ns, and 9 ns are shown respectively in Fig. 6a, Fig. 6c, and Fig. 8e for the turn-on and in Fig. 6b, Fig. 6d and Fig. 8f for the turn-off.

At turn-on, the  $I_S$  peak amplitude depends on the equivalent  $C_{OSS}$  of the GaN FETs in parallel and the  $C_{PCB}$ .  $I_{load}$  does not influence the peak because the high value of  $L_{diode} + L_{DB}$  (50 nH circa) lowers the  $I_{DB}$  slow rate.

Looking at the turn-on transient, with null delay, the  $I_S$  peak is minimized by splitting the total charge of the parasitic capacitances between GaN<sub>1</sub> and GaN<sub>2</sub> (Fig.6a). When GaN<sub>2</sub> has a delay, GaN<sub>1</sub> has a higher peak that lasts longer (Fig.6c). The area underneath the  $I_S$  curve is maximized for delay

equal to 9ns (Fig.6e) or longer. It represents the total charge of the parasitic capacitances. The current peak in Fig.6e is  $I_S=135$  A, which is only due to parasitic capacitances. It is the maximum achievable considering the number of GaN FETs mounted in the board and the designed PCB.

At turn-off, in the case of null delay, there is no current peak, and the  $I_S$  waveforms of GaN<sub>1</sub> and GaN<sub>2</sub> are congruent (Fig.6b). When GaN<sub>2</sub> turns off later than GaN<sub>1</sub>, GaN<sub>2</sub> shows an  $I_S$  peak (Fig.6d). In this example, the  $I_S$  turn-off peak of GaN<sub>2</sub> is maximum for delay=9 ns because the  $I_S$  reaches the amplitude of  $I_{load}$ . There is no contribution of the parasitic capacitances on the turn-off peak.

Overall, in a board with GaN FETs in parallel, the  $I_S$  peak due to the spread of parameters is maximum at turn-on in the condition of the low value of  $L_{diode} + L_{DB}$  (e.g., less than 2 nH circa in Fig.3a).

These results obtained with the board prototype featuring a high value of  $L_{diode} + L_{DB}$  are significant when it is intended to design a board with the same number of GaN FETs. Typically, the GaN FET-based inverter has a low value of  $L_{diode} + L_{DB}$  because another high-side device substitutes the freewheeling diode. In this scenario, the  $I_{DB}$  slew rate is steep enough to reach the  $I_{load}$  amplitude before the peak due to the parasitic capacitances.

The chosen EPC2302 GaN FET allows a pulse current of up to 400 A, while the parasitic capacitances of the parallel connection layout have been demonstrated to cause an  $I_S$  of up to 135 A. Hence, theoretically, a hypothetical inverter board designed in similarity to the tested prototype can work seamlessly even with extreme parameters spread, as in the case of this paper, with currents  $I_{load}$  up to 265 A.

## 4 Experimental evaluation of an inverter with four GaN FETs in parallel

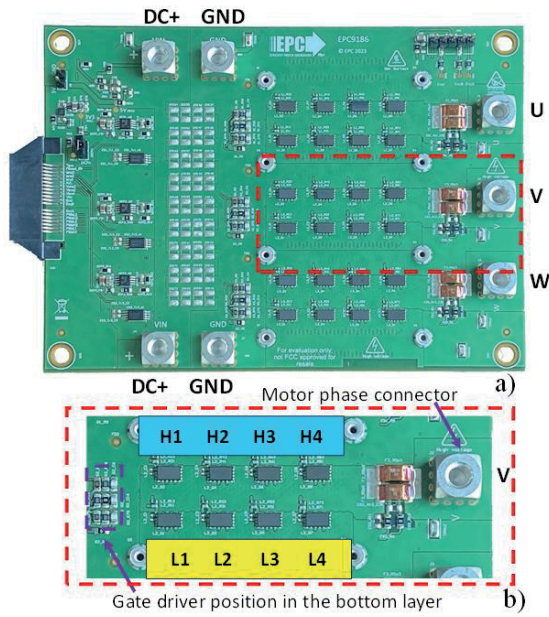
An inverter board with four parallel GaN FETs per each switching device has been designed using the experience from the 12 devices' parallel board characterization.

The top view of the experimental board picture is shown in Fig. 7a. In Fig. 7b, a half-bridge layout arrangement is highlighted.

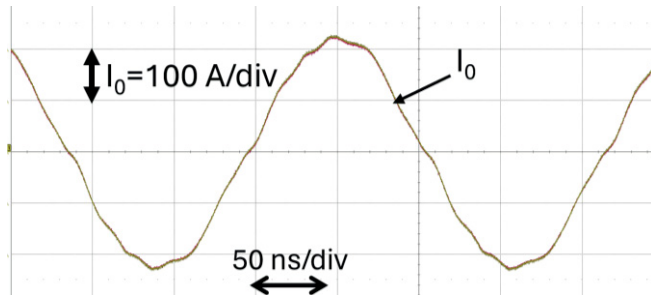
Each inverter leg is driven by a single gate driver that supplies the eight GaN FETs: four on the high side (H) and on the low side (L).

The inverter was tested by powering a 5 kW three-phase BLDC motor with a DC-Bus voltage of  $V_{DC}=60$  V. A DC-Bus capacitance of 720  $\mu$ F was used. The switching frequency was  $f_{sw}=20$  kHz.

The phase current  $I_0$  is measured and depicted in Fig. 8.



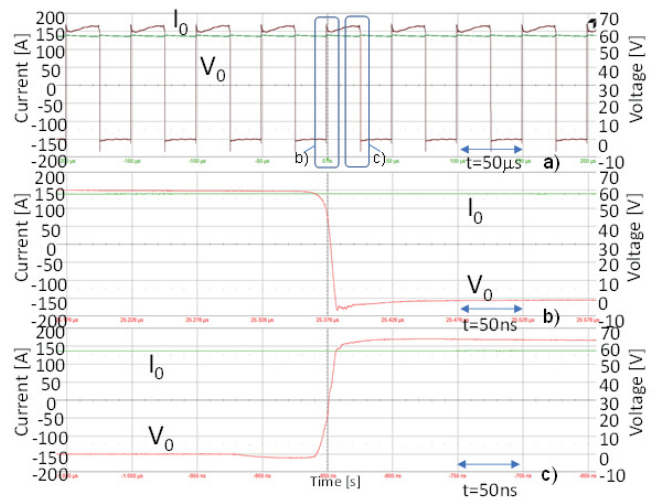
**Fig. 7** a) Top view of EPC9186 inverter board. b) Zoomed view of the half-bridge layout arrangement of phase V.



**Fig. 8**  $I_0=150\text{ A}_{\text{RMS}}$  output phase (U) current obtained using the inverter board with four GaN FETs in parallel per switch.  $f_{\text{sw}}=20\text{ kHz}$ .  $V_{\text{DC}}=60\text{ V}$ .

The experimental test shows that the designed inverter can supply the motor with a phase current of  $I_0=150\text{ A}_{\text{RMS}}$ .

Fig. 9a depicts the switching waveforms related to a motor phase at a current level of  $I_0=150\text{ A}_{\text{RMS}}$ . Fig. 9 b, and Fig. 9c depict zoomed views of the node voltage at turn-on and turn-off, respectively. Looking at Fig. 9, the switching voltage does not ring during commutations, and no overvoltage spikes are observed. This demonstrates that the layout's stray inductance is minimized and that the DC-link filter is well-designed.



**Fig. 9** a) Output phase (U) switching voltage  $V_0$  and current  $I_0$  waveforms.  $f_{\text{sw}}=20\text{ kHz}$ .  $V_{\text{DC}}=60\text{ V}$ ,  $I_0=150\text{ A}$ , b) zoomed view of the turn-on switching waveforms, c) zoomed view of the turn-off switching waveforms.

## 5 Conclusions

The paper investigates the advantages and issues of parallel-connected GaN FETs within inverter applications. The study discusses the effect of the spread of device parameters and layout considerations through a combination of simulation and practical experimentation through an explicitly developed board. This breakthrough promises to cater to the escalating demands of high-current motor drive applications, offering a more efficient and high-performance solution for the industry.

In simulations, the parameters spread, particularly the threshold voltage, has been revealed to cause a timing mismatch between the GaN FETs connected in parallel and, consequently, current peaks in the device that turns on earlier and turns off later. The strategy of separating the contribution of the parasitic capacitances to the rise in load current, as presented in this work, allows for empirical prediction of the maximum applicable load current level. Moreover, the design of the GaN FET-based inverter with parallel devices has followed the proposed analysis procedure and has demonstrated its capability to supply an electrical machine operating with a phase current of more than  $150\text{ A}_{\text{RMS}}$ .

Future studies should focus on thermal analysis and heat spreading issues when connecting in parallel GaN FETs with extreme parameters spread.

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